

Time-Reversal Symmetry and Majorana Correlators

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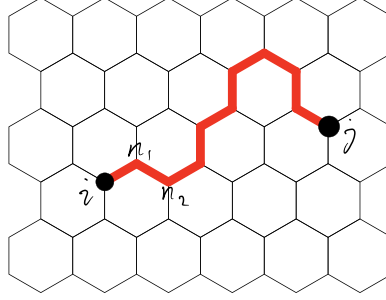


Figure 1: The configuration $\mathcal{M}$ contains only one single string. Points i and j are the end points of the sting, and $n_1, n_2, \dots$ are the points in the middle

Define the string operator

$$\mathfrak{S}_{\mathcal{M}} := \prod_{\langle kl \rangle} \sigma_k^{\alpha_{kl}} \sigma_l^{\alpha_{kl}}$$

and represent the Pauli algebra $[\sigma^\mu, \sigma^\nu] = 2i\epsilon^{\mu\nu\lambda}\sigma^\lambda$ using the Majorana operators in the way that $\sigma^\mu = \mathcal{P}_G i b^\mu c \mathcal{P}_G$ after the Gutzwiller projection, we then get

$$\mathfrak{S}_{\mathcal{M}} = \mathcal{P}_G \left(\prod_{\langle kl \rangle \in \mathcal{M}} \hat{u}_{kl} \right) \prod_{\langle kl \rangle \in \mathcal{M}} i c_k c_l \mathcal{P}_G = (-1)^{|\mathcal{M}|(|\mathcal{M}|-1)/2} i^{|\mathcal{M}|} \mathcal{P}_G c_i c_j \prod_{\langle kl \rangle \in \mathcal{M}} \hat{u}_{kl} \mathcal{P}_G$$

Where i and j are the end points of the string $\mathcal{M}$ illustrated in Fig.1. $|\mathcal{M}|$ stands for the length of the string.

The length of the string is related to the TRS property of the string operator. If $|\mathcal{M}|$ is an odd(even) number, then there will be an even(odd) number of middle points, i.e. points labeled by n in Fig.1, and then the string operator will be time-reversal even(odd). For $|\mathcal{M}|$ being odd, the simplest example is the single bond, which consists of two spin operators. For $|\mathcal{M}|$ being even, let's think of a sting consisting of two adjacent bonds, the string operator is then

$$(\sigma_i^x \sigma_{n_1}^x) (\sigma_{n_1}^y \sigma_j^y) = i \sigma_i^x \sigma_{n_1}^z \sigma_j^y$$

(w.o.l.g we have assumed $\alpha_{i,n_1} = x$ and $\alpha_{n_1,j} = y$) The ground state $|\psi\rangle$ of the Kitaev model is time reversal invariant $|\mathcal{T}\psi\rangle = |\psi\rangle$. So, the expectation value of any time-reversal odd quantity has to be zero. Therefore, the expectation value of the string operator with $|\mathcal{M}| = 2$ has to be zero

$$\langle (\sigma_i^x \sigma_{n_1}^x) (\sigma_{n_1}^y \sigma_j^y) \rangle = i \langle \sigma_i^x \sigma_{n_1}^z \sigma_j^y \rangle$$

because the operator $\sigma_i^x \sigma_{n_1}^z \sigma_j^y$ is time-reversal odd. Based on this logic one can get the corollary that *the expectation value of string operators with an even number of $|\mathcal{M}|$ must be zero in any state preserving TRS.*

Because $[\hat{u}_{ij}, \hat{H}] = 0$, we know the Kitaev ground state is an eigenvector of the gauge field operator $\hat{u}_{ij}$ after fixing the gauge. Denote by $|\psi_0\rangle = |u\rangle|M\rangle$ the Kitaev ground state with gauge field fixed ($|u\rangle$ can be arbitrary, it has not to be all-positive gauge). Then, the expectation value of the string operator can be evaluate in the

gauge-fixed Majorana space,

$$\langle \mathfrak{S}_{\mathcal{M}} \rangle = 0 = \underbrace{(-1)^{|\mathcal{M}|(|\mathcal{M}|-1)/2} i^{|\mathcal{M}|} \langle u | \left(\prod_{\langle kl \rangle \in \mathcal{M}} \hat{u}_{kl} \right) | u \rangle \langle M | c_i c_j | M \rangle}_{\neq 0}$$

So consequently, we then immediately get another result that $\langle c_i c_j \rangle = \langle M | c_i c_j | M \rangle$ in the Kitaev ground state must be zero if i and j are the end points of a string $\mathcal{M}$ with an even number of length $|\mathcal{M}|$.

In summary, the fact that the Majorana correlator $\langle c_i c_j \rangle$ is zero if i and j belong to the same type of sublattice (reflecting the string length to be even or odd) relies on only three conditions,

- TRS states (don't need to be the ground state)
- $[\hat{u}_{ij}, \hat{H}] = 0$.
- b -sector can be perfectly separated from c -sector in the theory.